Code

```python
import psycopg2

def manage_student_data():
    conn = None
    try:
        # 1. Connect to PostgreSQL

        conn = psycopg2.connect(dbname="college",user="postgres",password="sejal24post",host="localhost",port="5433")
        cur = conn.cursor()

        # 2. Create table if it doesn't exist
        cur.execute('''
            CREATE TABLE IF NOT EXISTS students (
                id SERIAL PRIMARY KEY,
                name VARCHAR(100),
                age INTEGER,
                grade VARCHAR(10)
            );
        ''')
        conn.commit()

        # 3. Take User Input
        print("\n--- Enter New Student ---")
        name = input("Name: ")
        age = int(input("Age: "))
        grade = input("Grade: ")

        # 4. Insert Data
        cur.execute("INSERT INTO students (name, age, grade) VALUES (%s, %s, %s)", (name, age, grade))
        conn.commit()
        print("Data saved successfully.\n")

        # 5. Fetch Data using fetchone()
        print("--- Current Student List ---")
        cur.execute("SELECT id, name, age, grade FROM students;")

        while True:
```

```python
        cur.execute("INSERT INTO students (name, age, grade) VALUES (%s, %s, %s)", (name, age, grade))
        conn.commit()
        print("Data saved successfully.\n")

        # 5. Fetch Data using fetchone()
        print("--- Current Student List ---")
        cur.execute("SELECT id, name, age, grade FROM students;")

        while True:
            row = cur.fetchone()
            if row is None:  # Break the loop if no more rows are found
                break
            print(f"ID: {row[0]} | Name: {row[1]} | Age: {row[2]} | Grade: {row[3]}")

    except (Exception, psycopg2.DatabaseError) as error:
        print(f"Error: {error}")

    finally:
        if conn is not None:
            cur.close()
            conn.close()
            print("\nDatabase connection closed.")

manage_student_data()
```

Creating Database and displaying the list of database

```
SQL Shell (psql)

psql (18.3)
WARNING: Console code page (437) differs from Windows code page (1252)
         8-bit characters might not work correctly. See psql reference
         page "Notes for Windows users" for details.
Type "help" for help.

postgres=# \l
                                                    List of databases
   Name    |  Owner   | Encoding | Locale Provider |      Collate       |       Ctype        | Locale | ICU Rules |   Access privileges
-----------+----------+----------+-----------------+--------------------+--------------------+--------+-----------+-----------------------
 demo      | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 postgres  | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 template0 | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           | =c/postgres          +
           |          |          |                 |                    |                    |        |           | postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           | =c/postgres          +
           |          |          |                 |                    |                    |        |           | postgres=CTc/postgres
(4 rows)


postgres=# create database college;
CREATE DATABASE
postgres=# \l
                                                    List of databases
   Name    |  Owner   | Encoding | Locale Provider |      Collate       |       Ctype        | Locale | ICU Rules |   Access privileges
-----------+----------+----------+-----------------+--------------------+--------------------+--------+-----------+-----------------------
 college   | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 demo      | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 postgres  | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 template0 | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           | =c/postgres          +
           |          |          |                 |                    |                    |        |           | postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           | =c/postgres          +
           |          |          |                 |                    |                    |        |           | postgres=CTc/postgres
(5 rows)


postgres=# \c college;
You are now connected to database "college" as user "postgres".
college=# \d
Did not find any relations.
college=#

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```

After Executing

Code output

```
(env) C:\Users\Dell\OneDrive\Desktop\TuteDude\Assignment\Assignment7>python student.py

--- Enter New Student ---
Name: Sejal
Age: 20
Grade: B
Data saved successfully.

--- Current Student List ---
ID: 1 | Name: Sejal | Age: 20 | Grade: B

Database connection closed.

(env) C:\Users\Dell\OneDrive\Desktop\TuteDude\Assignment\Assignment7>
```

Database output

```
SQL Shell (psql)
(4 rows)


postgres=# create database college;
CREATE DATABASE
postgres=# \l
                                                       List of databases
   Name    |  Owner   | Encoding | Locale Provider |      Collate       |       Ctype        | Locale | ICU Rules |   Access privileges
-----------+----------+----------+-----------------+--------------------+--------------------+--------+-----------+-----------------------
 college   | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 demo      | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 postgres  | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
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           |          |          |                 |                    |                    |        |           | postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           | =c/postgres          +
           |          |          |                 |                    |                    |        |           | postgres=CTc/postgres
(5 rows)


postgres=# \c college;
You are now connected to database "college" as user "postgres".
college=# \d
Did not find any relations.
college=# \d
               List of relations
 Schema |      Name       |   Type   |  Owner
--------+-----------------+----------+----------
 public | students        | table    | postgres
 public | students_id_seq | sequence | postgres
(2 rows)


college=# select * from students;
 id | name  | age | grade
----+-------+-----+-------
  1 | Sejal |  20 | B
(1 row)


college=#
```